

Assignment 1

Please submit the code and report (in pdf format) on Blackboard system before 23:59 March 29. Report can be written either in English or Chinese. Name your report as `studentID_Name.pdf`. **Use PySpark code for all tasks, unless specifically stated.**

Note:

- The report contains the screenshot of the program result (display first 5 lines). You can include the explanation of the key functions/codes in the report or in the code as comments.
- The pyspark code can be written in jupyter or .py files.

Q1

Here is a retail dataset. You can find the sale information from different shops in the world. Before you start to code, you may want to observe the data first, to see if there is any missing or abnormal data points.

(1) [1 point] Explicitly define a structured schema for the dataset and load the data. Remove any rows containing missing values (NULL/NA), and ensure that both **Quantity** and **UnitPrice** are strictly greater than 0. If **InvoiceNo** (invoice number) starts with the letter 'C', it indicates a cancelled/returned order. Please filter out all return orders.

(2) [1 point] Use **Linux Shell commands** to process the raw CSV file and find the **top 5 most frequently purchased product IDs (StockCode)** along with their occurrence counts.

(3) [2 point] Based on the cleaned dataset, what valuable business insights can you draw? (For example: What are the purchasing preferences in specific countries? During which time periods are orders most concentrated?) Please provide **two observations**, along with the corresponding PySpark code to support and visualize your conclusions.

Q2

OAS is a popular protein antibody sequence dataset that contains over one billion protein sequences. Here is a small subset of the unpaired protein sequences. Please extract the following four columns: `sequence`, `v_call`, `j_call`, and `cdr3_aa` from each data file. (you may want to remove the first line first). Remove the lines where the `cdr3_aa` length is strictly fewer than 8 or larger than 30. Ensure that the `cdr3_aa` column contains only valid uppercase letters corresponding to the 20 standard amino acids (**ACDEFGHIKLMNPQRSTVWY**). Discard any rows that contain unknown characters (such as **X**, *********), lowercase letters, or digits. Also, drop any rows where `v_call` is empty. Concatenate all valid data from the different files to output one final CSV file (use a comma `,` to separate, include only these four columns, and ensure there is only one header line at the top).

- (1) [1 points] Group by the (**v_call**, **j_call**) gene pair, count the occurrences of each V-J gene combination, and identify the longest sequence within each combination.
- (3) [1 points] For large files like `SRR12326775_1_Light_Bulk.csv`, we sometimes need to split it into chunks and process each chunk. Use `split` Linux shell command to equally split this file into 8 chunks, use for-loop in the shell script to [parallel process](#) each chunk. Then concatenate all chunk outputs into one file. In this small task, you only need to process the `SRR12326775_1_Light_Bulk.csv` file.
- (4) [1 points] Find the Top 3 most frequently occurring **v_call** genes in the dataset. For each of these 3 V genes, further calculate and output its most frequently paired **j_call** gene, along with the average **cdr3_aa** length for that specific V-J combination.

Q3

Suppose you are the CTO for a startup company and will design an international AI-power APP like 大众点评.

(1) [1 point] Use LLMs like chatgpt/doubao/kimi/etc. to design the system architecture of this product. Try **three** different prompt and then aggregate different answers for a better design.

(2) [2 points] Read the [DB-Engines Ranking](#). Compare the pros and cons among different databases of PostgreSQL , MongoDB , Snowflake . Which database (not limited to the above databases) will you select for your product? Explain your choice. Write prompts to use LLMs for this task.